

1. Fundamentals of Probability

- Event $A \Rightarrow 0 \leq P(A) \leq 1$
- $P(S) = 1$
- $P(\phi) = 0$
- $P(A \cup B) = P(A) + P(B) - P(A \cap B)$

A & B are mutually exclusive, $A \cap B = \phi$
 $P(A \cap B) = 0$

$\rightarrow P(A \cup B) = P(A) + P(B)$

Eg. 1 $1 \dots 6 \Rightarrow$ set of cards $\Rightarrow \{1, 2, \dots, 6\}$

$A \Rightarrow > 3 \Rightarrow \{4, 5, 6\} \Rightarrow P(A) = \frac{3}{6} = \frac{1}{2}$

$B \Rightarrow < 3 \Rightarrow \{1, 2\} \Rightarrow P(B) = \frac{2}{6} = \frac{1}{3}$

$C \Rightarrow$ event A & B are mutually exclusive
 $P(A \cup B) = P(A) + P(B)$

$\swarrow$
 $= \frac{1}{2} + \frac{1}{3} = \frac{3+2}{6} = \frac{5}{6}$

$\{1, 2\} \quad \{4, 5, 6\}$

$\rightarrow \underline{\{1, 2, 4, 5, 6\}}$

Eg. 2 a jar $\rightarrow$ 10 chocolates

$\swarrow$ 4 dark
 $\searrow$ 3 milk
3 white

$P(\text{dark} \cup \text{milk}) = \frac{4}{10} + \frac{3}{10} = \frac{7}{10}$ (mutually exclusive)

$$- \quad P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

eg. 3

$T \rightarrow 30$

$12 \rightarrow \text{drama } P(D) \quad D, M$

$10 \rightarrow \text{music } P(M)$

$5 \rightarrow \text{both } P(D \cap M)$

$P(D \cup M) = ?$

$$P(D \cup M) = P(D) + P(M) - P(D \cap M)$$

$$= \frac{12}{30} + \frac{10}{30} - \frac{5}{30} = \frac{17}{30}$$

$\Rightarrow 17$

eg. 4

$40\% \rightarrow \text{football}$

$30\% \rightarrow \text{basketball}$

$15\% \rightarrow \text{both}$

$$P(F) = 0.4$$

$$P(B) = 0.3$$

$$P(B \cap F) = 0.15$$

$$P(B \cup F) = ?$$

$$P(B \cup F) = 0.4 + 0.3 - 0.15$$

$$= 0.7 - 0.15 = 0.55 \Rightarrow 55\%$$

eg. 5

$45\% \rightarrow \text{blood test}$

$40\% \rightarrow \text{specialist}$

$65\% \rightarrow \text{either a blood test or a specialist}$

$P(\text{patients who needs both}) = ?$

$$\hookrightarrow P(B \cap S) = ?$$

$$P(B) = 0.45, \quad P(S) = 0.4, \quad P(B \cup S) = 0.65$$

$$P(B \cup S) = P(B) + P(S) - P(B \cap S)$$

$$0.65 = 0.45 + 0.4 - P(B \cap S)$$

$$P(B \cap S) = 0.45 - 0.65 = 0.2 \Rightarrow 20\%$$

2. Complements

- $P(A \cup B) = P(A) + P(B) - P(A \cap B)$

mutually exclusive events,

$$P(A \cup B) = P(A) + P(B)$$

- Complement of event A is the set of outcomes in the sample space S that are not in A .

→ A' or A^c

1. $A \cap A^c = \emptyset$ (no overlap)

2. $A \cup A^c = S$

3. $P(S) = 1$ $P(S) = P(A) + P(A') = 1$

$P(A') = 1 - P(A)$

4. $(A')' = A$

Eg. 1 $A \Rightarrow$ get at least 1 head when we toss 2 coins

$S = \{HH, HT, TH, TT\}$

→ $A = \{HT, TH, HH\}$
 $A' = \{TT\}$

→ $P(A) = \frac{3}{4}$

→ $P(A') = \frac{1}{4} = 1 - P(A)$

$$= 1 - \frac{3}{4} = \frac{1}{4}$$

Eg. 2 $A = \text{rolling an even no.} \Rightarrow P(A')$

$$S = \{1, 2, 3, 4, 5, 6\}$$

$$A = \{2, 4, 6\}$$

$$P(A) = \frac{3}{6} = \frac{1}{2}$$

$$P(A') = 1 - P(A) = 1 - \frac{1}{2} = \frac{1}{2}$$

$$A' = \{1, 3, 5\} \Rightarrow P(A') = \frac{3}{6} = \frac{1}{2}$$

Eg. 3 $\{1 \dots 10\}$ $A \Rightarrow$ draw a no. that is greater than 7.

$$A = \{8, 9, 10\} \Rightarrow P(A) = \frac{3}{10}$$

$$P(A') = 1 - P(A) = 1 - \frac{3}{10} = \frac{7}{10}$$

Eg. 4 5 red balls, 3 blue balls $\Rightarrow$ 8 w/o replacement
 $P(\text{we draw at least 1 red ball}) = ?$

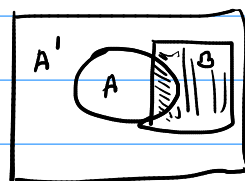
$\downarrow$ complement
 $P(\text{no red ball}) \Rightarrow P(\text{both balls being blue})$

$$P(\text{both blue}) = \frac{3}{8} \times \frac{2}{7} = \frac{3}{28}$$

$$P(\text{ }) = 1 - \frac{3}{28} = \frac{25}{28}$$

complements
in multiple
events

$$P(B) = P(\underline{A \cap B}) + P(A' \cap B)$$



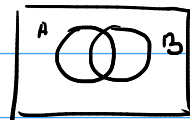
B happens either with A or w/o A.

Ex. 5

$$P(A \cap B') = 0.25, \quad P(A \cup B) = 0.70$$

$$\underline{\underline{P(B) = P(A) + 0.15}}, \quad P(A) = ?$$

$$\begin{aligned} P(A \cup B) &= P(A) + P(B) - P(A \cap B) \\ &= P(A) + P(A) + 0.15 - P(A \cap B) \\ &= 2P(A) + 0.15 - \underline{\underline{P(A \cap B)}} \end{aligned}$$



$$\begin{aligned} P(A) &= P(A \cap B) + P(A \cap B') \\ &= 2P(A) + 0.15 - (P(A) - P(A \cap B')) \end{aligned}$$

$$0.70 = P(A) + 0.15 + 0.25$$

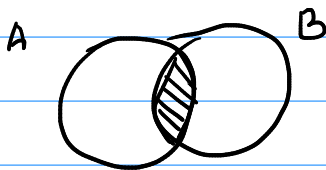
$$P(A) = 0.7 - 0.4 = 0.3$$

3. Venn Diagrams

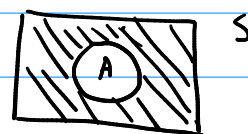
↳ a visual representation for events & their overlaps.

$\bigcirc \rightarrow \text{event}$,  (overlap)
intersections

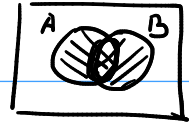
 $\rightarrow$ sample space S

 $A \cap B$ 
$$A \cup B$$


A'

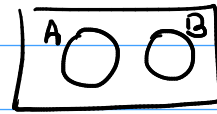


$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$



$$P(A \cap B) = \phi$$

$$P(A \cup B) = P(A) + P(B)$$

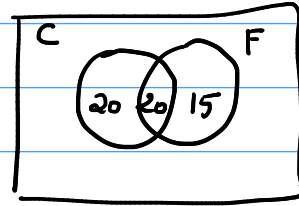


Eg. 1

40% $\rightarrow C$

35% $\rightarrow F$

20% $\rightarrow$ both

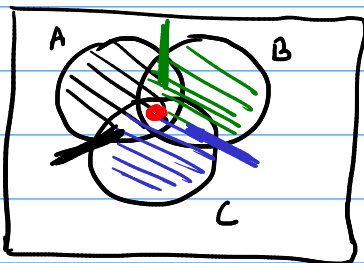


$$C \text{ or } F \Rightarrow C \cup F = 20 + 20 + 15 = 55\%$$

$$P(C \cup F) = 0.4 + 0.35 - 0.2 \quad //$$

$$= 0.75 - 0.2 = 0.55$$

- Events $\rightarrow A, B, \& C.$



$$P(A \cup B \cup C) = P(A) + P(B) + P(C)$$

$$- P(A \cap B) - P(B \cap C) - P(C \cap A)$$

$$+ P(A \cap B \cap C)$$

Eg. 2

50% $\rightarrow N$, 40% $\rightarrow C$, 35% $\rightarrow M$

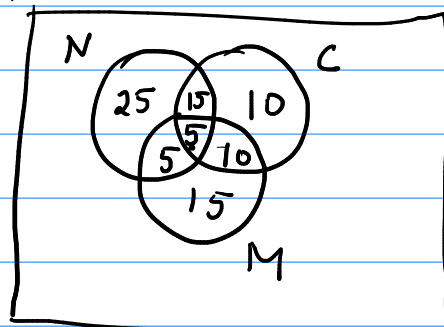
20% $\rightarrow N \& C$, 15% $\rightarrow C \& M$, 10% $\rightarrow N \& M$

5% $\rightarrow N \& C \& M$

percentage (none) = ? 15%

$$P(A \cup B \cup C) = 0.85 \Rightarrow 85\%$$

$$1 - 0.85 = 0.15 \Rightarrow 15\%$$



- Mutually exclusive events $\rightarrow$ events that cannot happen together ($P(A \cap B) = 0$)

Eg. 3 either take H or P, but never both.

$$0.6 \rightarrow P$$

$$0.4 \rightarrow H$$

$$P(P \cup H) = 0.6 + 0.4 = 1$$

Eg. 4

$$45\% \rightarrow P_i$$

$$30\% \rightarrow P_a$$

$$15\% \rightarrow \text{both}$$

Find % who like neither

$$P(P_i \cup P_a)'$$

$$\begin{aligned} P(P_i \cup P_a) &= P(P_i) + P(P_a) - P(P_i \cap P_a) \\ &= 0.45 + 0.3 - 0.15 = 0.75 - 0.15 \\ &= 0.6 \end{aligned}$$

$$\begin{aligned} P(P_i \cup P_a)' &= 1 - P(P_i \cup P_a) \\ &= 1 - 0.6 = 0.4 \Rightarrow 40\% \end{aligned}$$

Eg. 5

$$40\% \rightarrow D, 35\% \rightarrow C, 25\% \rightarrow B$$

$$10\% \rightarrow D \& C, 8\% \rightarrow C \& B, 6\% \rightarrow D \& B$$

$$3\% \rightarrow \text{all three}$$

% $\rightarrow$ who owns at least 1 pet.

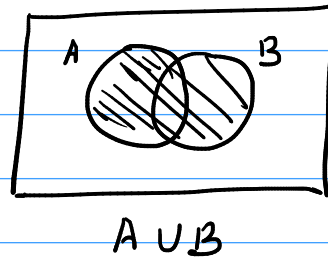
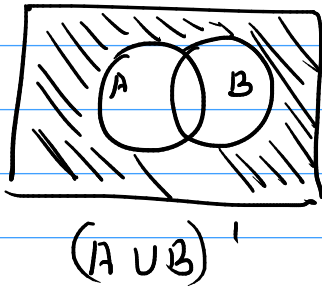
$$\begin{aligned} P(D \cup C \cup B) &= P(D) + P(C) + P(B) \\ &\quad - P(D \cap C) - P(C \cap B) - P(B \cap D) \\ &\quad + P(D \cap C \cap B) \\ &= 0.4 + 0.35 + 0.25 \\ &\quad - 0.1 - 0.08 - 0.06 \\ &\quad + 0.03 \\ &= 1 - 0.24 + 0.03 = 1 - 0.21 \\ &= 0.79 \Rightarrow 79\% \end{aligned}$$

4. de Morgan's laws

Or $\Rightarrow A \cup B \Rightarrow$ Union

And $\Rightarrow A \cap B \Rightarrow$ intersection

neither A nor B $\Rightarrow (A \cup B)' \Rightarrow$ complement of $(A \cup B)$



- $(A \cap B)' = A' \cup B'$
 - $(A \cup B)' = A' \cap B'$
- } When you take a complement, the union will turn into intersection & intersection into unions.
- $(A \cap B \cap C)' = A' \cup B' \cup C'$

Eg. 1

A $\Rightarrow$ action movies
B $\Rightarrow$ comedy

$$(A \cup B)' = A' \cap B'$$

neither like action nor they like comedy

70% $\rightarrow$ A, 50% $\rightarrow$ C, 30% $\rightarrow$ both, neither

$$\begin{aligned}
 P(A \cup B)' &= 1 - P(A \cup B) \\
 &= 1 - (P(A) + P(B) - P(A \cap B)) \\
 &= 1 - (0.7 + 0.5 - 0.3) \\
 &= 1 - (1.2 - 0.3) \\
 &= 1 - 0.9 = 0.1
 \end{aligned}$$

eg. 2

$$P(M \cup E)' = 0.25$$

$\nearrow$ master's degree
 $\nearrow$ exp > 5 yrs
 $\searrow$ 25% of people who neither have a master's degree nor they have a exp. of 5 years.

$$P(M) = 0.6$$

$$P(E) = ?$$

$$P(M \cap E) = 0.2$$

$$P(M \cup E) = 1 - P(M \cup E)' = 1 - 0.25 = 0.75$$

$$P(M \cup E) = P(M) + P(E) - P(M \cap E)$$

$$0.75 = 0.6 + P(E) - 0.2$$

$$= 0.4 + P(E)$$

$$P(E) = 0.75 - 0.4 = 0.35$$

- $(A \cup B)'$ or $(A \cap B)'$

eg. 3

$$P(A' \cup B' \cup C) = 0.65, \quad P(A \cap B \cap C) = 0.12$$

$$P(A \cap B) = ?$$

$$A \cap B \cap C' = (A' \cup B' \cup C)' = 1 - P(A' \cup B' \cup C)$$

$$= 1 - 0.65$$

$$= 0.35$$

$$P(A \cap B) = P(A \cap B \cap C) + P(A \cap B \cap C')$$

$$= 0.12 + 0.35 = 0.47$$

eg. 4

$$P((X \cup Y)') = 0.18, \quad P(X) = 0.5, \quad P(X \cap Y) = 0.2, \quad P(Y) = ?$$

$$P(X \cup Y) = 1 - P((X \cup Y)')$$

$$= 1 - 0.18$$

$$= 0.82 = P(X) + P(Y) - P(X \cap Y)$$

$$= 0.5 + P(Y) - 0.2 = 0.3 + P(Y)$$

$$P(Y) = 0.82 - 0.3 = 0.52$$

eg. 5 $P(A' \cap B') = 0.40$, $P(A) = 0.45$, $P(B) = 0.5$
 $P(A \cap B) = ?$

$$(A' \cap B') = (A \cup B)' = 0.4$$

$$P(A \cup B) = 1 - P(A \cup B)'$$

$$= 1 - 0.4 = 0.6$$

$$0.6 = 0.45 + 0.5 - P(A \cap B)$$

$$0.6 = 0.95 - P(A \cap B)$$

$$P(A \cap B) = 0.95 - 0.6 = 0.35$$

5. Inclusion - Exclusion Principle

40% $\rightarrow T$

50% $\rightarrow C$

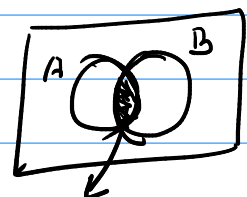
20% $\rightarrow$ both

$$0.4 + 0.5 = 0.9$$

$$0.9 - 0.2 = 0.7$$

$\rightarrow$ prevents the problem of double counting

$$- P(A \cup B) = P(A) + P(B) - P(A \cap B)$$



$$- P(A \cup B \cup C) = P(A) + P(B) + P(C) - P(A \cap B) - P(B \cap C) - P(C \cap A) + P(A \cap B \cap C)$$

$$- P(A \cup B \cup C \cup D) = \sum P(\text{singles}) - \sum P(\text{pairs}) + \sum P(\text{triplets}) - P(A \cap B \cap C \cap D)$$

Eg. 1 45% $\rightarrow$ E, 35% $\rightarrow$ F, 20% $\rightarrow$ both
% of those who know at one lang.

$$P(E) = 0.45, P(F) = 0.35, P(E \cap F) = 0.2$$

$$\begin{aligned} P(E \cup F) &= P(E) + P(F) - P(E \cap F) \\ &= 0.45 + 0.35 - 0.2 \\ &= 0.8 - 0.2 = 0.6 \Rightarrow 60\% \end{aligned}$$

Eg. 2 50% $\rightarrow$ S, 40% $\rightarrow$ D, 35% $\rightarrow$ M
20% $\rightarrow$ S & D, 15% $\rightarrow$ D & M, 10% $\rightarrow$ S & M
5% $\rightarrow$ S & D & M

% of people who does at least 1 activity.

$$\begin{aligned} P(S \cup D \cup M) &= P(S) + P(D) + P(M) - P(S \cap D) - \\ &\quad P(D \cap M) - P(S \cap M) + P(S \cap D \cap M) \\ &= 0.5 + 0.4 + 0.35 - 0.2 - 0.15 - 0.1 + 0.05 \\ &= 1.3 - 0.45 = 0.85 \end{aligned}$$

85% people do at least 1 activity.

Eg. 3 30% $\rightarrow$ L, 25% $\rightarrow$ T, 40% $\rightarrow$ S, 20% $\rightarrow$ W
12% $\rightarrow$ L & T, 18% $\rightarrow$ L & S, 10% $\rightarrow$ L & W, 15% $\rightarrow$ T & S,
8% $\rightarrow$ T & W, 12% $\rightarrow$ S & W
6% $\rightarrow$ L & T & S, 4% $\rightarrow$ L & T & W, 5% $\rightarrow$ T & S & W
7% $\rightarrow$ L & S & W
2% $\rightarrow$ L & T & S & W

% of people who does own any gadget.

$$P(L \cup T \cup S \cup W) = \sum P(\text{singles}) - \sum P(\text{pairs}) + \sum P(\text{triplets}) - P(L \cap T \cap S \cap W)$$

$$\sum p(\text{singles}) = 0.3 + 0.25 + 0.4 + 0.2 = 1.15$$

$$\sum p(\text{pairs}) = 0.12 + 0.18 + 0.1 + 0.15 + 0.08 + 0.12 = 0.75$$

$$\sum p(\text{triplets}) = 0.06 + 0.04 + 0.05 + 0.07 = 0.22$$

$$P(L \cap T \cap S \cap W) = 0.02$$

$$P(L \cup T \cup S \cup W) = 1.15 - 0.75 + 0.22 - 0.02$$
$$= 0.4 + 0.2 = 0.6$$

$$P(L \cup T \cup S \cup W)' = 1 - P(L \cup T \cup S \cup W)$$
$$= 1 - 0.6 = 0.4$$

40% of people who does not own any gadget.

Q. 25% $\rightarrow C$, 20% $\rightarrow F$, 15% $\rightarrow B$, 10% $\rightarrow$ all three
5% $\rightarrow C \& F$, 4% $\rightarrow F \& B$, 3% $\rightarrow C \& B$

% of people who plays none of these games.

$$P(C \cup F \cup B)' = P(C' \cap F' \cap B') = ?$$